

Biopolymers, and the computational methods used to study them

A bibliometric mapping of the field, with a Latin American and Colombian focus

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Abstract

This report maps the scientific literature on biopolymers and measures how much of it uses computational chemistry, chemoinformatics or artificial intelligence. It then narrows to Latin America and Colombia. The corpus is built from OpenAlex and cross-checked against Lens, and the method layer is classified by reading abstracts rather than by matching keywords. Two results shape everything that follows. Keyword bibliometrics overstates the computational share of the field worldwide, because the vocabulary of biopolymers overlaps that of structural biology; and it understates the share in Latin America, because much regional work is invisible to affiliation matching. Reading the abstracts reverses the ranking of contexts that the keyword counts produce.

1 What this report answers

Three questions, in the order the work addressed them.

1. Where and when is research on biopolymers being done, and by whom?
2. How much of that research uses computational chemistry, chemoinformatics or artificial intelligence, and which methods on which polymers?
3. What does the Latin American and Colombian position look like inside that map?

The thematic scope is deliberately wide: every biopolymer recognised as a subject of scientific study, anchored on the lines this group works in jointly, namely cellulose and nanocellulose, polyhydroxyalkanoates, and lignin.

2 Method

2.1 How the corpus was defined

The core corpus is the field that *identifies itself* as biopolymers: works whose title or abstract carries a biopolymer framing term. This was reached empirically. Blocks for each polymer family were probed first, and every family block intersected with a biopolymer framing turned out to be a subset of the generic string, so the generic string is the union rather than a narrowing.

Two acronym tests decided the wording of every query. The bare acronym PHA returns 56,015 records against 8,422 when bound to its expanded form, and bare DFT 471 against 295 inside the core corpus. Bare acronyms are used nowhere.

Every search string is a versioned file that is never edited once run, so a count can always be traced to the exact text that produced it.

2.2 Sources

Scopus and Web of Science are not used: the institution has no subscription. The triangulation therefore runs on OpenAlex as the base corpus and Lens as an independent index, with Semantic Scholar and Europe PMC recovering abstracts that OpenAlex lacks.

This turned out to be more informative than a comparison of two commercial indexes would have been. Table 1 shows Lens holding most of the world corpus but a far smaller share of the regional one.

Table 1: OpenAlex against Lens, same concept, same window. Lens covers the world corpus well and the regional corpora poorly, so the choice of source matters most exactly where this report focuses.

Block	Region	OpenAlex	Lens	Lens as % of OpenAlex
Core field	World	142,254	129,729	91.2
Core field	Latin America	8,147	5,412	66.4
Core field	Colombia	598	342	57.2
Method layer	World	3,912	3,323	84.9
Method layer	Latin America	211	104	49.3
Method layer	Colombia	17	4	23.5
Cellulose	Latin America	14,222	9,813	69.0
Cellulose	Colombia	897	558	62.2
PHA/PHB	World	22,225	20,241	91.1
PHA/PHB	Latin America	1,444	1,037	71.8
PHA/PHB	Colombia	88	64	72.7
Lignin	Latin America	11,464	8,598	75.0
Lignin	Colombia	810	635	78.4

The overlap is partial in both directions. Across the matched corpora, between 76% and 100% of Lens records are also in OpenAlex, but only between 24% and 75% of OpenAlex records are in Lens. Lens still contributes works OpenAlex lacks, so the union of the two is larger than either. A single-source count of Latin American output is therefore a lower bound, and this is a result rather than a caveat.

2.3 Reading the abstracts

A keyword query cannot tell a molecular dynamics study of a cellulose nanocrystal composite from a molecular dynamics study of haemoglobin allostery, yet only the first is biopolymer materials science. Every abstract in the method layer was therefore classified by a language model: whether the work is materials research or biological-function research, which polymer family, which computational method, and whether experiments are reported. Every call is cached on a hash of the prompt, so the classification is reproducible and re-running it is free.

One correction was applied after inspecting the output rather than trusting it. The first pass offered a catch-all computational category, and it was absorbing factorial designs, response-surface methodology and curve fitting to the authors’ own measurements. That inflated the Colombian computational share to 12.5% against a defensible 3%. The ambiguous bucket alone was re-asked with a question naming the distinction, and statistical design of experiments is now excluded

from every computational count. The first-pass classifications are kept on disk so the correction is auditable.

2.4 Validating the screening

A classification produced by a language model is worth nothing without an agreement figure. 125 abstracts were double-coded by a second, independent reader on a different model, blind to the first coder’s labels. Asking the same model to check its own work would have it repeat its own mistakes and report high agreement with itself.

Agreement is reported as Cohen’s kappa rather than percent agreement, because percent agreement flatters any classification with a dominant class.

Table 2: Agreement between the two independent coders on the random sample.

Dimension	Agreement	Cohen’s κ
materials vs biological	94.0%	0.878
computational at all	92.0%	0.782
method family	88.0%	0.816
polymer family	80.0%	0.767

The distinction the whole correction depends on, materials research against biological function, reaches the highest agreement of the four. On the conventional reading, 0.61–0.80 is substantial and above 0.80 almost perfect.

The two coders also disagree in a consistent direction. On the random sample of 100, the first coder called 79 works computational against the second coder’s 73, a relative over-detection of 8.2%. Five of the seven disputed works were labelled machine learning, and reading them shows why: they are reviews and experimental papers that mention artificial intelligence as a closing perspective without performing any computation.

Every screened share in this report is therefore an upper bound, and the strict world figure of 1.28% is closer to 1.18% once the bias is applied. This runs the same way as every other correction found here.

3 Results

3.1 Size, growth and geography

Table 3: The field by context. Growth rates use complete years only. The method share here is the *keyword* estimate, corrected in Section 3.2.

Context	Works, all years	2015–2025	Share of world	Growth/yr	Open access
World	145,743	90,931	100.00%	11.4%	43.2%
Ibero-America	13,095	9,445	10.39%	13.7%	58.5%
Latin America	8,162	6,035	6.64%	14.7%	55.5%
Colombia	598	465	0.51%	27.0%	66.7%

The field holds 145,743 works and grows at 11.4% a year. Both regional contexts grow faster than the world, Latin America at 14.7% and Colombia at 27.0%, so the region is closing the relative gap rather than falling behind. Colombia’s rate is computed on a small base and says more about the base than about capacity. Regional output is markedly more open: 66.7% of Colombian work is open access against 43.2% worldwide.

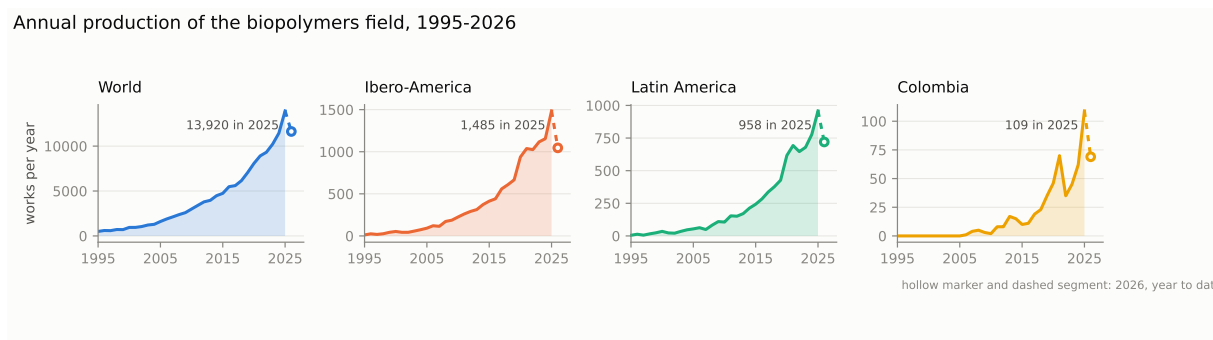


Figure 1: Annual production by context. Each panel carries its own scale, since the contexts differ by three orders of magnitude. The hollow marker and dashed segment are 2026, a partial year.

The top three countries, United States, China and India, hold 50,777 works between them, just over a third of the field. Only one Latin American country reaches the top twenty: Brazil.

Inside Latin America the order by volume is Brazil (4,450), Mexico (1,559), Argentina (794), Colombia (598), Chile (492), Ecuador (232).

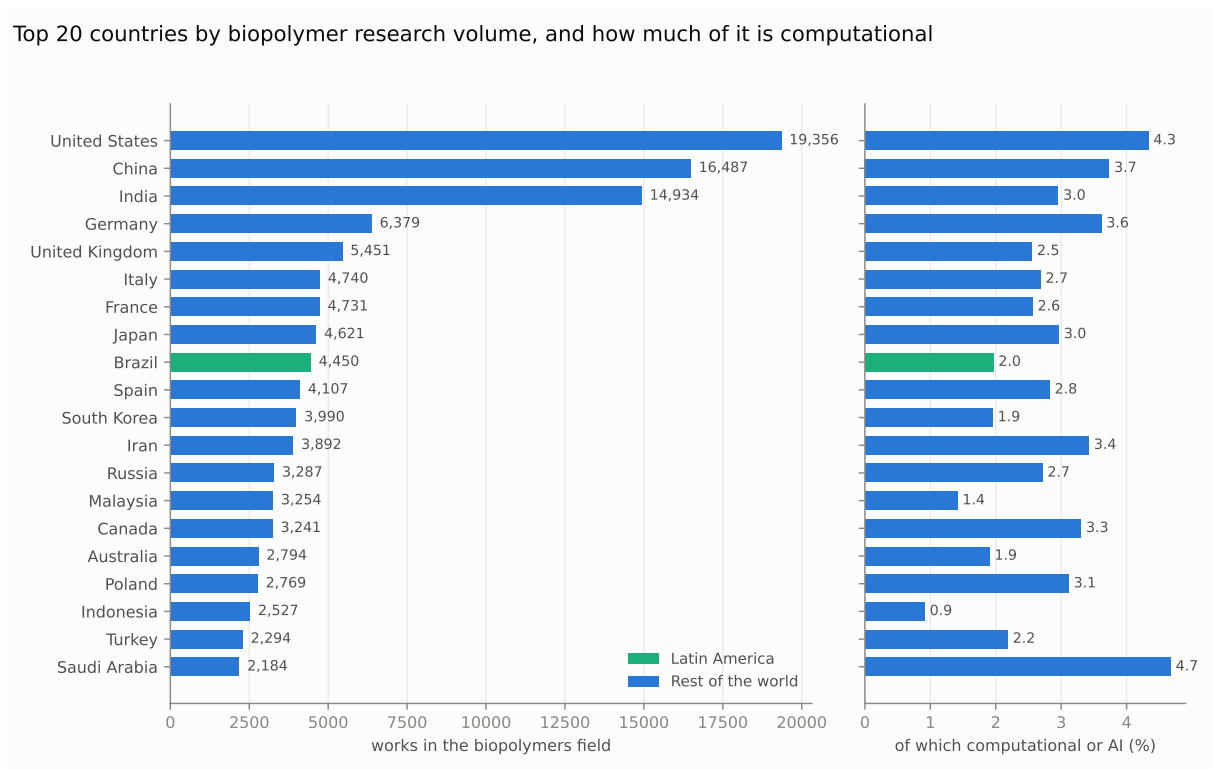


Figure 2: The twenty largest producers, and how much of each country's output is computational. Volume and intensity are separate panels; they are different measures and do not share an axis.

3.2 The computational and artificial-intelligence layer

This is the central result. The keyword estimate puts 2.63% of the field on some computational, chemoinformatic or AI method. Reading the abstracts shows only 56.6% of that layer to be biopolymer materials research; the rest is protein folding, enzyme mechanism and nucleic-acid biology, pulled in because the framing terms overlap. Restricting to materials research brings the share to 1.49%, and restricting further to works with a genuine computational method gives 1.28%.

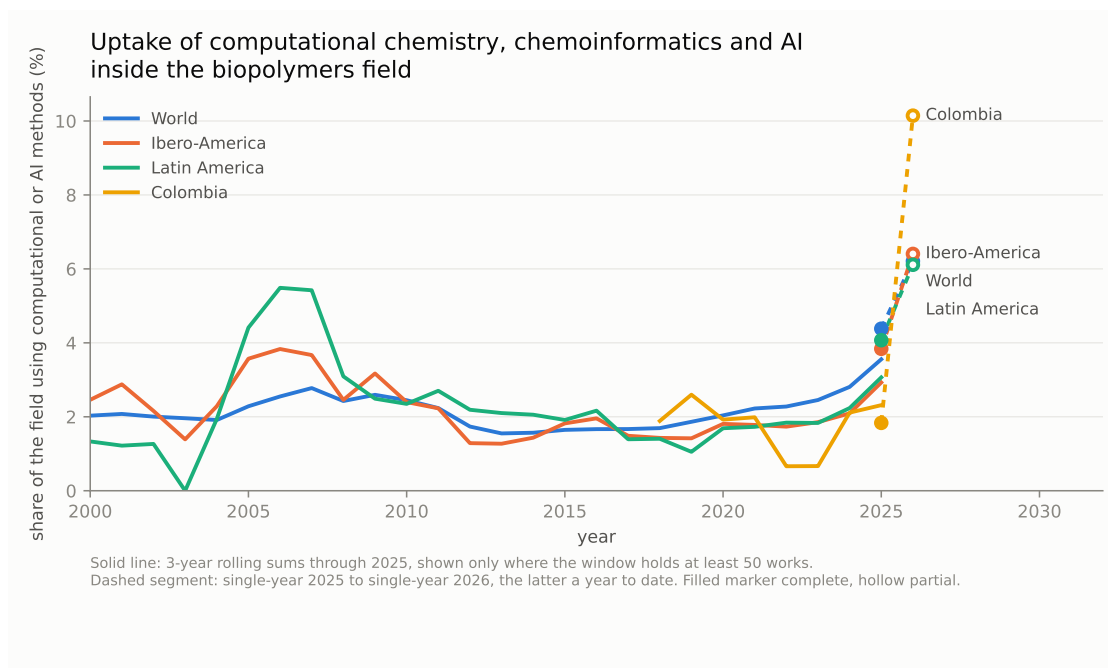


Figure 3: Uptake of computational methods inside the field. The solid line is a three-year rolling share; the dashed segment joins single-year 2025 to single-year 2026, the latter a year to date. Ratios on small denominators are suppressed.

The partial year is not dropped. 2026 is incomplete and is kept, flagged, because it carries the strongest signal in the data.

Table 4: 2026 to date against the whole of 2025. The field as a whole is well short of its 2025 volume at the same date, while the method layer has already passed it in every context.

Context	Whole field			Method layer		
	2025	2026 YTD	%	2025	2026 YTD	%
World	13,920	11,629	84	610	722	118
Ibero-America	1,485	1,046	70	57	67	118
Latin America	958	720	75	39	44	113
Colombia	109	69	63	2	7	350

3.3 Keyword counts understate the region

Table 5: Screening the whole regional field, rather than the keyword-selected method layer, finds computational work the query missed.

Corpus	Materials works	With a method	Screened share	Keyword share
Colombia, whole field	446	20	4.48%	1.94%
Latin America, whole field	5,751	197	3.43%	2.20%

Reading abstracts raises the regional estimate substantially, and it changes the ordering: Colombia sits *above* the Latin American average, not below it as the keyword counts say. The keyword-based ordering should not be quoted.

Part of the reason is mechanical. In the world method corpus, 12.7% of authorships match no institution entity at all, so the country they belong to receives no credit. A 2025 paper with two

Peruvian affiliations and one Mexican one counts, in the data, as Mexican alone. Since every regional corpus here was selected on institution country, works whose regional affiliations all went unmatched never entered it. Per-country loss is worst for the smallest producers. The regional counts in this report are a lower bound.

3.4 Which methods, on which polymers

Table 6: Method mix of the materials-only subset, as a percentage of each context. The world leans on molecular dynamics; the region leans on machine learning and quantum chemistry.

Method	World	Ibero-America	Latin America	Colombia
Molecular dynamics	33.0	28.2	24.2	22.2
Machine learning	24.8	27.5	29.3	22.2
Quantum chemistry	16.5	22.8	24.2	33.3
Docking	5.7	4.7	4.0	0.0
Solubility modelling	3.5	2.0	0.0	0.0
QSAR / QSPR	1.4	0.0	0.0	0.0
Generative	0.6	0.0	0.0	0.0
Other molecular modelling	0.7	2.0	3.0	11.1
Other informatics	0.2	0.7	1.0	0.0
Statistical design of experiments	0.2	0.0	0.0	0.0
None detected	13.4	12.1	14.1	11.1

Generative methods and QSAR are absent from the region entirely, and worldwide the generative slice is a handful of works. That is the emptiest part of the map and the most obvious opening.

3.5 The regional deficit is partly linguistic

Every count so far came from English queries. Spanish and Portuguese term variants were run against the same regional slices and the set difference taken, so what is reported is not how many records the Spanish and Portuguese strings return but how many they return that the English string does *not*.

Table 7: What Spanish and Portuguese terms add. The last column is the set difference: records the Spanish and Portuguese strings return that the English string does not.

Block	Region	English	Spanish/Portuguese	Added
Core field	Latin America	8,159	1,364	1,252 (+15.3%)
Core field	Colombia	598	183	157 (+26.2%)
Cellulose	Latin America	14,669	4,218	3,952 (+26.9%)
Cellulose	Colombia	902	303	281 (+31.1%)
PHA/PHB	Latin America	1,444	221	199 (+13.8%)
PHA/PHB	Colombia	88	36	30 (+34.1%)
Lignin	Latin America	11,527	1,815	1,686 (+14.6%)
Lignin	Colombia	811	158	145 (+17.9%)

The addition is substantial and it is largest for Colombia. A bibliometric count of Latin American biopolymer research built on English terms alone misses between a seventh and a third of the corpus, depending on the slice. Taken with the affiliation-matching loss of Section 3.2, the regional figures in this report understate the region twice over, for two independent and separately measurable reasons.

3.6 The three anchored lines

Table 8: The biopolymer lines this group works in jointly.

Line	World	Latin America	Colombia	Computational, world	Computational, region
Cellulose and nanocellulose	378,840	14,683	902	1.64%	1.69%
PHA / PHB	22,378	1,444	88	1.30%	1.52%
Lignin	181,260	11,534	811	2.08%	1.86%

Lignin is the most computationally studied of the three and PHA the least. Latin America holds a larger share of world output in PHA and lignin than in cellulose, which fits a region working from agro-industrial residues.

The three anchored biopolymer lines

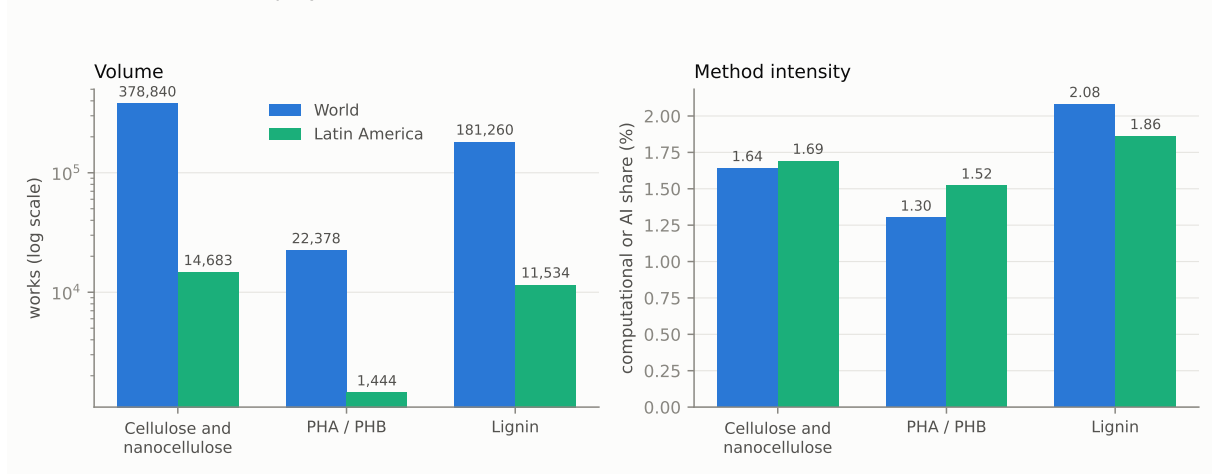


Figure 4: Volume and method intensity of the three anchored lines.

3.7 Collaboration and community structure

Table 9: Collaboration, counted per work.

Corpus	Works	International	Two or more Latin American countries
World, method layer	3,328	30.4%	0.6%
Latin America, whole field	8,175	34.4%	6.0%
Colombia, whole field	598	44.1%	23.9%
Latin America, method layer	211	45.5%	9.0%

Colombia is the most regionally connected corpus measured. Nearly 24% of Colombian works involve a second Latin American country, against 6% for the region as a whole. Brazil's strongest ties run outward instead, to the United States, Portugal and Spain, and dwarf its ties inside the region.

There is no standing computational community. Classic Lotka behaviour gives an exponent near 2. Every corpus here is steeper, and the method layers are steepest of all: of the 14,861 authors in the world method layer, 86.3% appear exactly once, with a fitted exponent of 3.46. The layer is made of one-off contributions. That is what a field looks like before a speciality forms, not after.

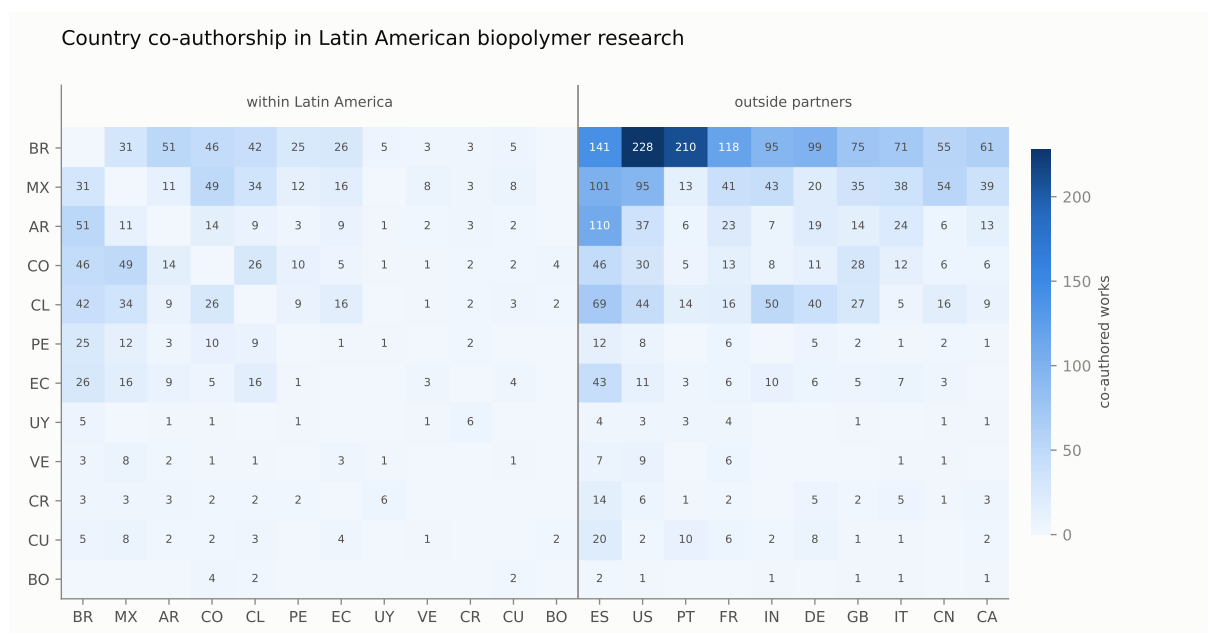


Figure 5: Country co-authorship in Latin American biopolymer research, shown as a matrix. A node-link drawing of the same 115-node graph is a hairball; the network files are provided for VOSviewer instead.

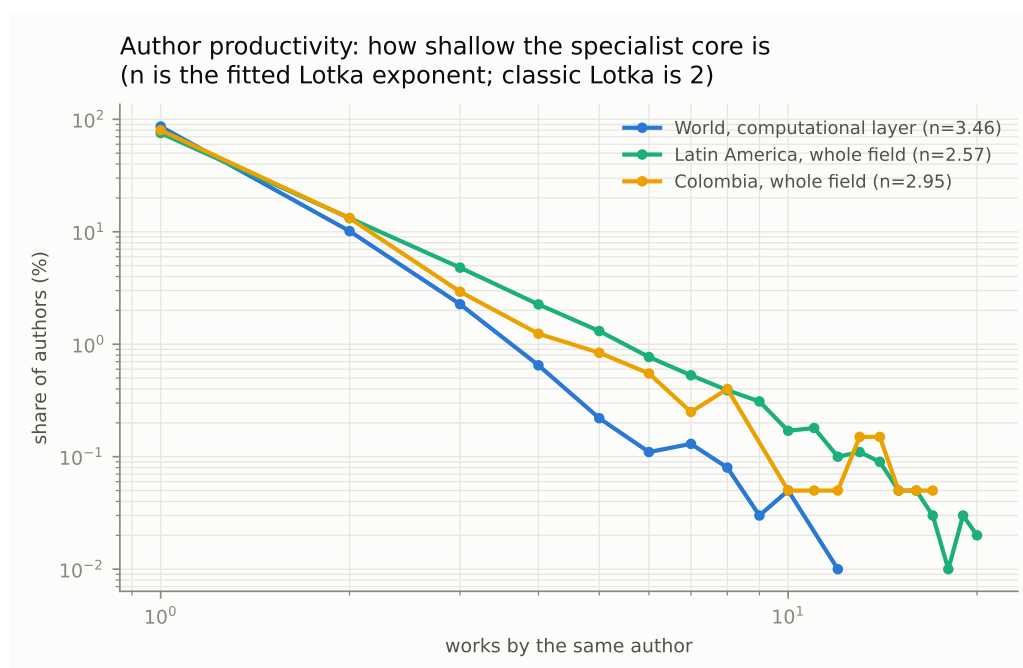


Figure 6: Author productivity. A steeper slope means a shallower core of specialists.

And it has no home venue. Bradford’s core zone is the set of journals carrying the first third of a corpus. Worldwide it is 2.4% of journals, which is normal. For the Latin American method layer it takes 13.2% of its journals to reach the first third. That work is scattered, with no venue acting as a centre.

3.8 Colombia

The Colombian institutional ranking is led by Universidad Nacional de Colombia (96), Universidad del Valle (66), University of Cartagena (47), Universidad de Antioquia (47). Universidad de Sucre appears with 25 works.

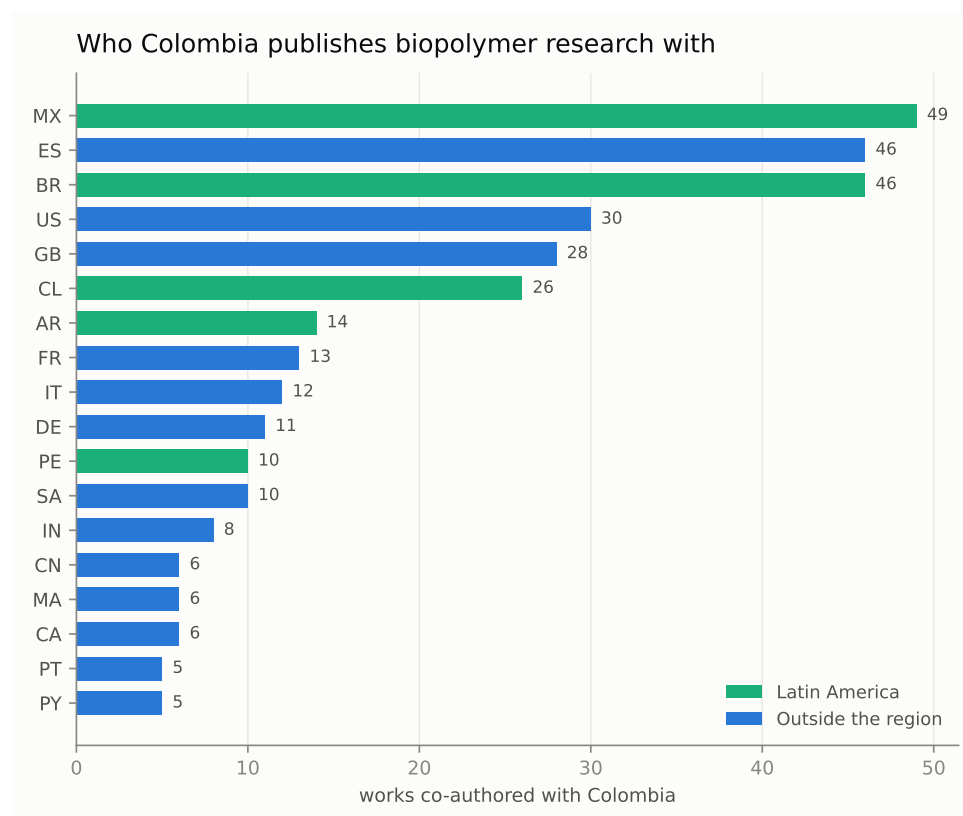


Figure 7: Colombia’s co-authorship partners, split between the region and outside it.

4 What the map shows

1. **The computational layer is thin and it is accelerating.** Around one work in eighty in the biopolymers field uses a genuine computational method on a material. That share roughly doubled over a decade and 2026 is running ahead of 2025 in every context measured.
2. **The regional deficit is volume, not method.** Latin America and Colombia adopt computational methods at rates comparable to, and by the screened measure above, the world. What the region lacks is scale and continuity of output, not access to computational practice. This reverses the usual framing and it changes what a recommendation should say.
3. **The method mix differs by region.** The world leans on molecular dynamics; Latin America leans on machine learning; Colombia leans on quantum chemistry. An infrastructural reading is available, since sustained dynamics needs computing time that the other

two need less of, but the Colombian denominator is small and this belongs in the discussion as a hypothesis rather than in the results as a finding.

4. **Generative methods are essentially absent**, worldwide and entirely so in the region. Of everything in this report, that is the clearest opening.
5. **There is no community and no venue**. Most authors in the computational layer publish there once, and the regional slice has no core journals. A thesis that wants to change something has a structural target, not only a topical one.

5 Limitations

- **Two sources, not four**. Scopus and Web of Science were unavailable. OpenAlex and Lens disagree by about a third on regional counts, which bounds how precise any regional figure here can be.
- **Affiliation matching loses regional work**. Quantified in Section 3.2. The regional counts are a lower bound.
- **Abstract coverage**. Roughly 30% of the method-layer records carry no abstract in OpenAlex. Semantic Scholar and Europe PMC recovered 4,070 of them; the rest could not be read and are excluded from every screened share.
- **The screening is not yet validated**. A hand-coded sample double-coded against the automatic classification would give an agreement figure. Until that exists, the screened shares should be reported as what they are, a machine reading.
- **English-language queries**. Regional work published in Spanish and Portuguese is undercounted.
- **Country attribution counts a work once per country present**, so country columns sum to more than the corpus.

6 What to do next

1. Hand-code a validation sample against the screening and report the agreement.
2. Add Spanish and Portuguese term variants and measure what they add.
3. Read the generative and large-language-model works one by one. There are few enough to read them all, and they are the frontier this thesis is pointing at.
4. Take the affiliation-matching loss seriously as a result in its own right. It is a measurable, publishable statement about how invisible small Latin American institutions are to the infrastructure that counts science.